Child labor
Child labor was cheap, and in demand, as small hands could reach into machines. These children spinning cotton in South Carolina, in 1903, have bare feet, because nails on their shoes might produce a spark, causing a fire.

Forging a revolution
These men working in a Minnesota metalworks formed part of a large but unprotected workforce. By the late 19th century, Minnesota was one of the largest producers of iron ore in the US.

made by James Watt (see below) have remained an essential element of steam engines ever since. When, in the 1770s, Watt went into business with Matthew Boulton, an entrepreneur and factory owner, they manufactured steam engines to Watt's patented design. These went on to power all stages of industrial production—pumping the mines; powering machinery in the factories and mills; and driving the steamships and railroad locomotives.

The "iron horse"

The huge demand for steam resulted in an increased need for coal. Improvements in both the mining of coal and its distribution via canals, and later, railroads, dramatically cut its cost. The use of refined coal (coke) to smelt iron (see BEFORE) further fueled industrialization by enabling engineers to build better tools and machines. Iron was also used as a building material for bridges, ships, and railroads. By 1850, about 2 million tons of iron had been used for railroad tracks and there were 6,214 miles (10,000 km) of train tracks around Britain. Known as "iron horses," locomotives could pull huge loads and reach speeds of up to 40 mph (65 km/h). The speed and efficiency of the railroads made the growth of the great manufacturing cities possible, and by the 1840s, it had cut the cost of moving goods by up to 50 percent. Railroad timetables changed timekeeping, with the standardized use of Greenwich Mean Time (GMT) replacing local time across Britain.

Steam powered
Trains, like this British one from 1908, were both the product and driving force of the Industrial Revolution. They were enabled by the growth of coal mining, and transported raw materials and goods at high speed.

Dark satanic mills

With the harnessing of steam power, factories and mills no longer needed to be sited near natural resources such as rivers, and became increasingly concentrated in towns. Textile production rapidly mechanized; by 1835 there were more than 120,000 power looms in textile mills. "Domestic" or "cottage" work (where home-workers were paid per item produced), ran alongside factory work, where workers operated machines that carried out just one task in a production line.

Reaction to progress

The Industrial Revolution undoubtedly improved productivity, and drove technological and economic progress, but it also became synonymous with appalling living and working conditions. Men, women, and children flocked to the cities, but with so many seeking employment, the value of their labor was reduced, and they worked long hours for low pay. Many formed trade unions, but workers' conditions improved only slowly: legislation passed was limited in scope, and frequently ignored by factory owners.

Opposition to industrialization also came from skilled workers, who had been made obsolete by mechanization, and unemployed factory workers. Rioting, and the wrecking of »

TRADE UNION **An organization devoted to protecting the interests of its members, who are drawn from a specific profession or trade.**

IDEA

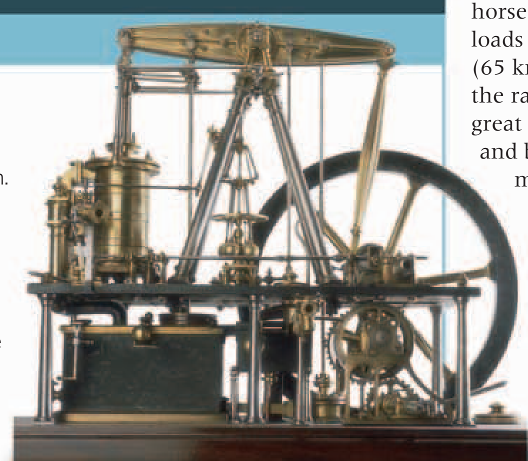
FREE ENTERPRISE CAPITALISM

Free enterprise capitalism works on the basis that all of industry (property, business, factories, and transport) should be privately owned, and that the products of industry should be sold in a free market, through a system of supply and demand. For this to happen, industry has to be unrestricted by government control, subsidies, or tariffs. Growing populations in the 18th and 19th centuries provided a larger market for new products, and steady growth in wages increased buying power, helping to satisfy the laws of supply and demand.

EARLY FRENCH BANKNOTE

INVENTION

THE STEAM ENGINE



In 1712 Thomas Newcomen built an "atmospheric engine" in which a vacuum produced by condensing steam caused atmospheric pressure to pull down a beam. This device was useful for pumping water out of coal mines. Sixty years later James Watt improved its efficiency, and added modifications for driving machinery.
In 1804 Richard Trevithick put a high-pressure steam engine on wheels to make the first steam locomotive, and in 1830 the *Rocket* pulled the first passenger train on George Stephenson's Liverpool–Manchester railroad.